MIL-C-4116B
9 DECEMBER 1974
SUPERSEDING
MIL-C-4116A
10 February 1954

MILITARY SPECIFICATION

CONTAINERS; SHIPPING, REUSABLE, WOOD, AIRCRAFT ENGINES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for one type of demountable, reusable, plywood sheathed shipping container and interior supports for aircraft engines and accessories.

1.2 Classification. Shipping containers shall be of the following styles as specified (see 6.2).

Style I - Vertical mount - So designed that the engine is mounted in the container with its longitudinal axis perpendicular to the base of the container.

Style II - Horizontal mount - So designed that the engine is mounted in the container with its longitudinal axis parallel to the base of the container.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

NN-P-530	Plywood, Flat Panel
QQ-A-250/2	Aluminum Alloy 3003, Plate and Sheet
QQ-S-698	Steel, Sheet and Strip, Low-Carbon
TT-E-485	Enamel, Semi-gloss, Rust-Inhibiting
TT-I-558	Ink, Marking Stencil, Opaque, for Nonporous Surfaces (Metal, Glass, Etc.)
TT-P-636	Primer Coating, Alkyd, Wood and Ferrous Metal
TT-W-546	Wood-Preservative; Acid Copper Chromate Mixture
TT-W-549	Wood-Preservative; Ammonical Copper Arsenate Mixture
TT-W-550	Wood-Preservative; Chromated Copper Arsenate Mixture
TT-W-571	Wood-Preservative, Treating Practice

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Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5607	Engine, Gas Turbine, Preparation for Storage and Shipment of, Process for
MIL-E-6058	Engines, Aircraft Reciprocating, Preparation for Shipment and Storage of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-731	Quality of Wood Members for Containers and Pallets

Federal

FED-STD-595	Colors
FED TEST METHOD STD NO. 101	Preservation, Packaging, and Packaging Materials: Test Procedures

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

No. 343	Specifications and Standards Applicable to Aircraft Engines and Propellers, Use of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article tests. This specification makes provisions for first article tests.

3.2 Materials.

3.2.1 Wood. Wood shall be either group II, III, or IV as classified in MIL-STD-731, unless otherwise specified. All framing members, cleats, blocking, etc, shall be free from loose knots and from knots having a diameter greater than 1/4 the width of the face, and free from cross grain diverging more than 1 inch in 10 inches along the grain.

3.2.2 Plywood. All plywood shall conform to NN-P-530, type I.

3.2.2.1 Metal edging for plywood sheathing. The edges of plywood sheathing shall be covered with hot-rolled sheet steel not less than 0.030 inch

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thick conforming to QQ-S-698. The metal edging shall be of "U-shaped" form and shall overlap both sides of the plywood sufficiently to be securely fastened to the plywood. Both faces of the metal edging shall be cleaned and treated with one coat of olive-drab primer conforming to TT-P-636 before placement on the plywood.

3.2.3 Selection of materials. Specifications and standards for all materials, parts, and government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall, except as provided in the following paragraph, be selected in accordance with ANA Bulletin No. 343.

3.2.3.1 Standard parts. Standard parts (MS and AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS and AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and construction. The container shall consist of individual side, end, base, top panels, and collapsible engine and accessory supports, so designed that when disassembled, the six panels and interior supports may be bundled with steel straps to occupy a minimum space for shipment or storage. Each container assembled, as for shipment, shall be able to support two like containers with contents enclosed.

3.3.1 Engine and accessory supports. The container shall be so designed that engine or accessory mountings are readily accessible, and shall permit the installation and removal of the engine and accessories from the container by means of overhead lifting devices, including opening and closing the container through a sequence of simple operations. The mounts shall be designed to permit complete enclosure of the engine in a water vapor-proof barrier. (Information as to the type of barrier to be accommodated may be obtained from the procuring activity.) Interior supports shall include all parts and fastenings necessary to receive and secure the engine within the container. When specified (see 6.2) internal supports for designated accessories shall be provided.

3.3.2 Size. The containers shall be of a minimum size which will provide clearance between the engine and the interior of the container with full allowance for movement of the engine as permitted by the cushioning medium or supports.

3.3.3 Skids. Two or more hardwood skids, one piece or laminated, shall be provided on the bottom of the container. Steel bolts and washers of a size adequate to prevent splitting, shall be installed transversely near each end of the skids. The vertical clearance between the bottom

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plane of the skids and the bottom of the container shall be a minimum of 4 inches. Skids shall be of group IV woods, as specified in MIL-STD-731. The wood shall be sound and free from all defects that would materially affect strength or interfere with fastenings. The wood shall be free from knots, having a diameter greater than 1/4 the width of the face, and from cross grain diverging more than 1 inch in 10 inches along the grain. The maximum moisture content at the time of assembly shall be 50 percent. The lower 1/2 of the ends of the skids shall be beveled 45 degrees from the horizontal. The attachment of the skids to the container shall be designed to a total strength of the assembly (of all skids) in longitudinal shear equal to 10 times the gross weight of the container. The skids shall be dimensioned and so mounted with respect to the center of gravity of the loaded container as to permit transportation on handling trucks 42 x 100 inches in dimension without additional blocking or supports. (Conformance to last requirement shall be determined by dimensional check.) After fabrication, drilling of holes, scarfing ends, etc, has been completed, the skids shall be preserved in accordance with TT-W-571. Waterborne preservatives conforming to TT-W-550, TT-W-549, or TT-W-546 are recommended in the stated order of preference.

3.3.4 Lifting. The container shall be so constructed that is may be lifted by forklift trucks from the side and end without employing additional blocking, and without permanent deformation to the container. Where the gross weight of the loaded container is 5,000 pounds or less, the maximum length of forks shall be taken as 36 inches; over 5,000 pounds, 40 inches. Where the container skids run parallel to the long axis of the container, two openings, 3 inches high by 10 inches wide, minimum, spaced approximately 30 inches center to center, or one 3 inch by 40 inch opening, minimum, shall be provided. The opening, or openings, shall be symmetrically disposed about the center of balance of the loaded container.

3.3.4.1 Lifting devices. External lifting devices such as hooks, rings, or eyes, shall be provided at the four corners of the top of the container. Additional devices shall be provided when necessary to facilitate handling. The lifting devices shall be capable of withstanding the tests prescribed herein without failure or visible distortion. Hooks, if provided, shall be of such size permitting the use of 1 inch cable slings. Rings or eyes shall have a minimum inside diameter of 2-3/4 inches. All devices shall be so recessed that no part extends in excess of 1/8 inch above the surface of the container when not in use. Positive, reusable means shall be provided for securing the devices within the recess.

3.3.5 Ventilation. Four 1 inch diameter openings, cleaned of all splinters, shall be bored in each end of the container, spaced 2-1/2 inches apart on center in the form of a square. The direction of the holes from the outside shall be at an angle of 45 degrees upward. The clusters of holes shall be located not less than 10 inches nor more than 18 inches below the top surface of the container and shall be equidistant from the sides of the container. Each cluster of holes shall be enclosed on the inside of the container by a wood or plywood baffle plate placed 2 inches from the openings and wood filler strips along the sides and bottom of the baffle plate.

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The bottom filler shall be placed level with the bottom of the lower two holes in order that any water entering the enclosure will drain out. Prefabricated metal ventilators, approved by the procuring activity, may be substituted.

3.3.6 Drainage. Two 1/2-inch diameter holes, cleaned of all splinters, shall be bored through the bottom of the container in any area where water may be trapped.

3.3.7 Inspection port. In one end of the container, an inspection port conforming to figure 1 shall be provided. The opening shall be not less than 7 inches in diameter and shall be covered by a top-hinged lid. The part shall be so located as to be in alignment with the window in the engine-protective envelope. (Details of engine-protective envelopes showing window locations may be obtained from the procuring activity.)

3.3.8 Engine record receptacle. A suitable receptacle for engine records shall be provided. This may be integral with the inspection port cover (see figure 1) or may be a flat, weatherproof structure 9 inches by 1 inch by 1/2 inch or a metal moistureproof cylindrical receptacle, 4 inches minimum inside diameter and 11 inches minimum inside length, with a removable cover which is secured to the inside of the container adjacent to the inspection port in such manner as to be accessible through the inspection port.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.4.1 Major sections of the container which may be otherwise assembled in more than one relative position shall be furnished with assembly guide(s) and suitable markings which will permit assembly in one relative position only.

3.5 Finish.

3.5.1 Container. All exterior and interior surfaces of the container shall be finished with enamel conforming to TT-E-485, color - olive drab, number 34087 of FED-STD-595.

3.5.2 Supports. All metal parts of engine supports and accessory supports shall be treated with one coat of primer conforming to TT-P-636, followed by a finish coat of enamel conforming to TT-E-485, color - olive drab, number 34087 of FED-STD-595.

3.6 Fastenings. Container closure fastenings and engine and accessory support fastenings shall be such that only common hand tools will be required for assembly or disassembly.

3.7 Performance. When tested in accordance with the requirements of section 4, the container, interior mounts, and container accessories shall reveal no significant structural weaknesses. No deformation shall have occurred which will not permit the ready disassembly and reassembly of

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the container. The contents shall be undamaged, and there shall be no evidence of a substantial amount of shifting of the contents within the container which would create conditions likely to cause damage during shipment. Unless otherwise specified, examination of the condition of engines used as test loads shall be conducted by the activity responsible for approval.

3.8 Markings. The following information shall be marked on each container with waterproof white ink, color number 37875 of FED-STD-595, and conforming to TT-I-558. Markings shall be at the locations and of the sizes as indicated.

a. On each side and end, near the top, in 2 inch letters: TOP

b. On each side and end, in 1 inch letters and below "TOP:"
LIFTING HOOKS ON TOP

c. On each side and end, in 2 inch letters:
REUSABLE CONTAINER - DO NOT DESTROY

d. On each side near the bottom at the center of balance of the loaded container, a vertical line 6 inches long and 1 inch wide, and adjacent thereto in 1 inch letters:
CENTER OF BALANCE

e. On one end, adjacent to the inspection port, and to the engine record receptacle in 1 inch letters:
INSPECTION PORT - ENGINE RECORD INSIDE (as applicable)

f. On each side:

CAUTION . . . . . (4 inches high)
AIRCRAFT ENGINE . . . . . (3 inches high)
DO NOT DROP . . . . . (3 inches high)

3.8.1 The interior surface of each container component, and each engine support or accessory support assembly, shall be marked with the model designation of the engines for which the container is designed.

3.9 Identification of product.

3.9.1 Nameplate. A nameplate, permanently and legibly marked with the following information, including all information required to be inserted in the blanks indicated, shall be permanently attached to each major section of the container. The nameplate shall be fabricated from aluminum conforming to QQ-A-250/2, temper O.

CONTAINER; SHIPPING, REUSABLE, WOOD, AIRCRAFT ENGINE
Specification MIL-C-4116B
Serial No. (To be assigned as directed by the procuring activity)
Manufacturer's Part No.

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Contract or Order No.
Manufacturer's Name or Trademark
National Stock No.
The abbreviation "US"

3.10 Installation instructions. Each container shall be accompanied by one set of instructions printed on durable paper contained in an envelope placed in the engine record receptacle. Instructions shall include a step-by-step procedure for installing and removing the engine model, or models, for which the container is designed. The instructions shall be approved by the procuring activity prior to printing.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of containers shall be classified as follows:

a. First article tests (see 4.3.1).

b. Quality conformance tests (see 4.4).

4.3 First article tests.

4.3.1 First article test samples. First article test samples shall consist of one container upon which approval is desired. Samples shall be appropriately identified. The container shall be examined as specified by the procuring activity and tested in accordance with paragraph 4.6.1 through 4.6.5. The test results shall be approved by the procuring activity before production is commenced.

4.4 Quality conformance tests. These tests shall consist of the examination of the product in accordance with paragraph 4.6.1.

4.4.1 Rejection and retest. Failure of any container to conform to any one of the requirements of this specification shall be cause for rejection of the container. Containers which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to eliminate the cause of the defects found in the original shall be furnished to the inspector.

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4.5 Test conditions. For all first article tests, the container shall be loaded with the engine and designated accessories for which it is designed, fully assembled and in a prepared-for-shipment condition, provided that the engine and designated accessories are made available by the procuring activity. When the procuring activity has not made a live engine available for tests required by paragraph 4.3.1, interim approval may be granted for the engine container prior to the shipping test, based on the results of tests conducted with an appropriate dummy engine. A shipment test with the dummy engine will not be required for this interim approval. Final approval of the engine container will be withheld until the live engine test has been performed satisfactorily.

4.6 Test methods.

4.6.1 Examination of product. Each container submitted for acceptance under contract shall be carefully examined to determine conformance with the materials, design, finish, and workmanship requirements of this specification and with any applicable drawings.

4.6.2 Dynamic tests. Each container shall be loaded (see 4.5) and subjected to the following tests as applicable and in accordance with Federal Test Method Standard No. 101:

a. Cornerwise drop (rotational) test, method 5005.

b. Drop test (free fall), method 5007.

c. Edgewise drop (rotational) test, method 5008.

d. Roll-over test, method 5014.

e. Tip-over test, method 5018.

f. Pendulum-impact test, method 5012, or incline-impact test, method 5023.

g. Vibration (repetitive shock) test, method 5019.

4.6.3 Static loading.

4.6.3.1 Concentrated load resistance. A load, equal to two times the weight of the container carrying the contents for which it is designed, shall be applied to the top of the container in a manner simulating the effect of two similar containers being stacked on top of the container.

4.6.3.2 Distributed load resistance. A load sufficient to insure a bearing pressure of 175 psf shall be evenly applied to the top surface of the container. This test shall not be applied to areas with curved or dome-shaped cross section.

4.6.4 Handling characteristics test.

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4.6.4.1 Assembly and disassembly. The engine shall be placed in the container and the container shall be made ready for shipment. It shall then be removed from the container. Records shall be maintained of the tools required for this work.

4.6.4.2 Lifting test. The loaded container shall be lifted clear of the ground by a forklift truck with forks spaced 30 inches center to center and transported for a distance of not less than 100 feet. The container shall then be deposited, the fingers of the truck placed between the skids of one end of the container, the container lifted free of the ground and again transported for a distance of 100 feet. If the location of the center of gravity does not permit the free lift of the container from one end, using fork extensions, the one end shall be raised to a height of 6 inches by the forklift and the container drawn along a concrete or similarly smooth surface for a distance of not less than 100 feet. If the loaded container weighs 5,000 pounds, or less, the forks shall be approximately 36 inches long; over 5,000 pounds, 40 inches. Observations concerning stability of the container on the forks and any tendency of the forks to distort or puncture the containers shall be recorded.

4.6.5 Shipping test. The shipping test shall consist of the shipment of an engine in operating condition, shipped in a container designed for the given engine, for a distance of not less than 2,500 miles by rail freight, loaded in a car equipped with AAR standard trucks. Preparation shall be made by taking an engine in good condition, replacing all main bearings with new parts and preparing for shipment as specified in MIL-E-5607 or MIL-E-6058. Following the shipment, the engine shall be disassembled, the bearings shall be inspected critically for brinelling, cracking, and other signs of undue wear, and reinstalled. The engine shall then be operated to determine any defects attributable to the rail shipment. A full report of this test shall be submitted in triplicate to the procuring activity. An alternate test to the 2,500 mile rail shipment is method 5019, vibration (repetitive shock) of Federal Test Method Standard No. 101.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Bolts, washers, nuts, or other unpainted assembly or accessory hardware, or unpainted metal container parts, shall be treated with a preservative conforming to one of the specifications listed in MIL-P-116. All loose parts shall be packed in a wood box which shall be securely attached to the bundle or, if packing in boxes is impractical, loose parts shall be securely attached to or packed within the bundle.

5.2 Packing. Each container shall be disassembled and arranged in a compact bundle secured with steel strapping in such manner as to insure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery. Bundles shall be capable of withstanding storage, rehandling, and reshipment without the necessity of rebundling.

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5.3 Marking of shipments. Each container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The containers described in this specification are suitable as standby items for use in the shipment of gas turbine aircraft engines and horizontally mounted piston type aircraft engines when material for metal containers conforming to MIL-C-5584 may be critical.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of container required.

c. Size of container required, specify in order of length by width by depth.

d. Style of shipping container (see 1.2).

e. Internal supports for accessories (see 3.3.1).

f. Container marking if different from 3.8.

g. Additional marking if required (see 3.8, 3.9, and 5.3).

h. Packing instructions if different from 5.2.

i. When first article test sample is required (see 4.3.1).

6.3 Vibration design. The provisions for vibration isolation of the engine from shipping conditions should adequately protect major engine components from damage that will seriously affect engine operation. In order to accomplish this, the designer should provide maximum isolation in the vertical and horizontal transverse directions relative to the main axis (coincident with the main shaft) of the engine for style II containers, and in vertical axis (coincident with the main shaft) for style I containers. The frequency range to be particularly guarded against is 2.5 to 7.5 cps with a 1 inch double amplitude between 2.5 and 5 cps and amplitudes corresponding to 1g acceleration between 5 and 7.5 cps. Every effort should be made to avoid sharp resonant frequencies (amplification factors in excess of 10) in these ranges. Consideration should also be given to minimizing, if practicable, amplification of vibrations between 7.5 and 55 cps and amplitudes corresponding to 1g. It is generally safe to assume, for design purposes, that frequencies over 55 cps will be damped out by skid and container structure although frequencies as high as 1,000 cps with 10g or more acceleration have been reported in the literature.

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6.4 Design alternates. While the requirements of this specification are based on the anticipation that a separate, flexible, water vapor-proof barrier may be applied around the contents, it is not the intent of this specification to exclude such designs as may incorporate other means of providing a water vapor-proof barrier. The right is reserved, therefore, to accept for first article tests, or to receive bids from manufacturers of, such containers as include a separate water vapor-proof barrier or such containers as are constituted of panel materials which provide an airtight enclosure when the container is assembled and closed by normal, practicable means.

CUSTODIAN:

Navy - AS

Air Force - 69

Reviewer Activity:

Air Force - 70, 71, 80, 82, 84

User Activity:

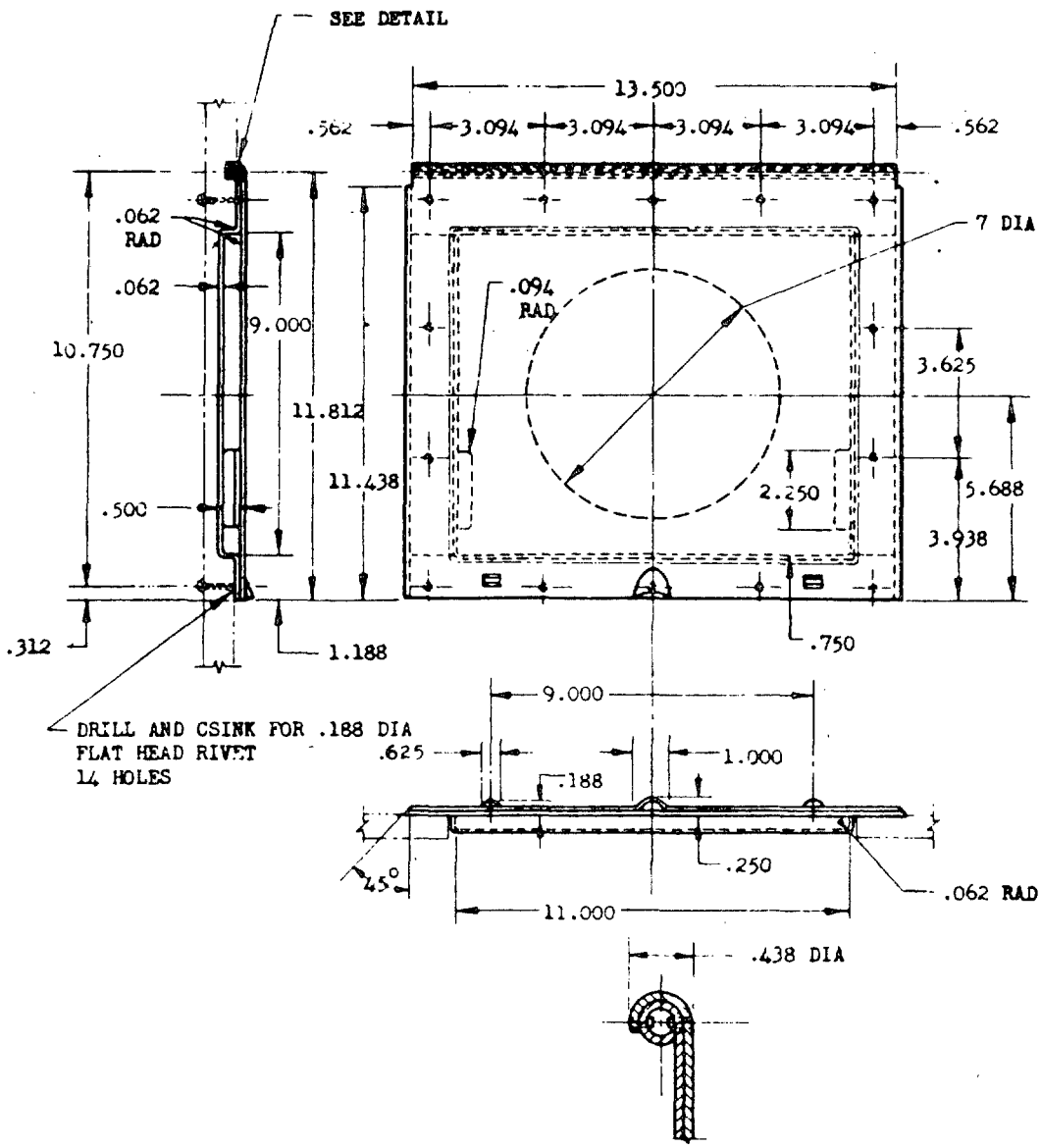
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.010, ANGLES ±1°.

LOG DATA AND INSPECTION PORT.

FIGURE 1. Inspection port

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE
MIL-A-54003A - ARRESTER, SPARK, M-1941

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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